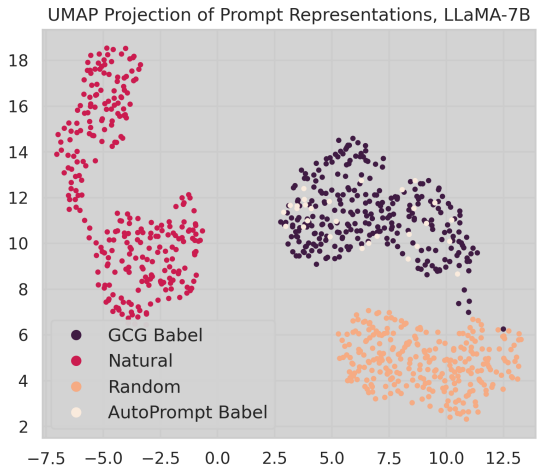

Figure 13. U-map of the last hidden state representations of LLaMA2-7B for successful Babel generated with GCG and AutoPrompt algorithms, natural and random prompts.

lower, with a 15% success rate compared to 40% for GCG, prompts generated by both methods appear in the same cluster in the model's representation space, as illustrated in our UMAP analysis in Figure 13.

B.7. Distance Metric Results

In addition to the exact match and conditional perplexity metrics used throughout the paper, we conducted additional experiments using a distance metric between the target text and the text generated by the model when prompted with gibberish input to measure the success rate of manipulation. In particular, we employed the mxbai-embed-large-v1 sentence embedding model trained using AnglE loss. Interestingly, we observed that most generations either have perfect cosine similarity with the target text, or it falls below 0.7, indicating that most successful generations are reproduced verbatim, as can be seen in the histogram in Figure 14. Also, in Table 9, we report the success rates of Babel prompts, where the success rate is measured as the percentage of generations with a cosine similarity between the target text embedding and the generation embedding above 0.9. It can be observed that the distance-based metrics are highly correlated with the exact match rate metrics.

Table 8. Success rate of Babel prompts measured using distance metric.

Model	Wikipedia	CC-News	AESLC	AdvBench
Vicuna 7B	80%	47%	40%	83%
Vicuna 13B	84%	54%	47%	80%
LLaMA 7B	51%	32%	27%	60%
LLaMA 13B	33%	14%	16%	44%

B.8. Prompting Details

We use the prompt templates for LLaMA and Vicuna models provided through the FastChat platform (Zheng et al., 2023). In particular, we use an empty system message and the conversation template includes gibberish prompt within *[INST]* tags for LLaMA models and after *USER:* token for Vicuna models.

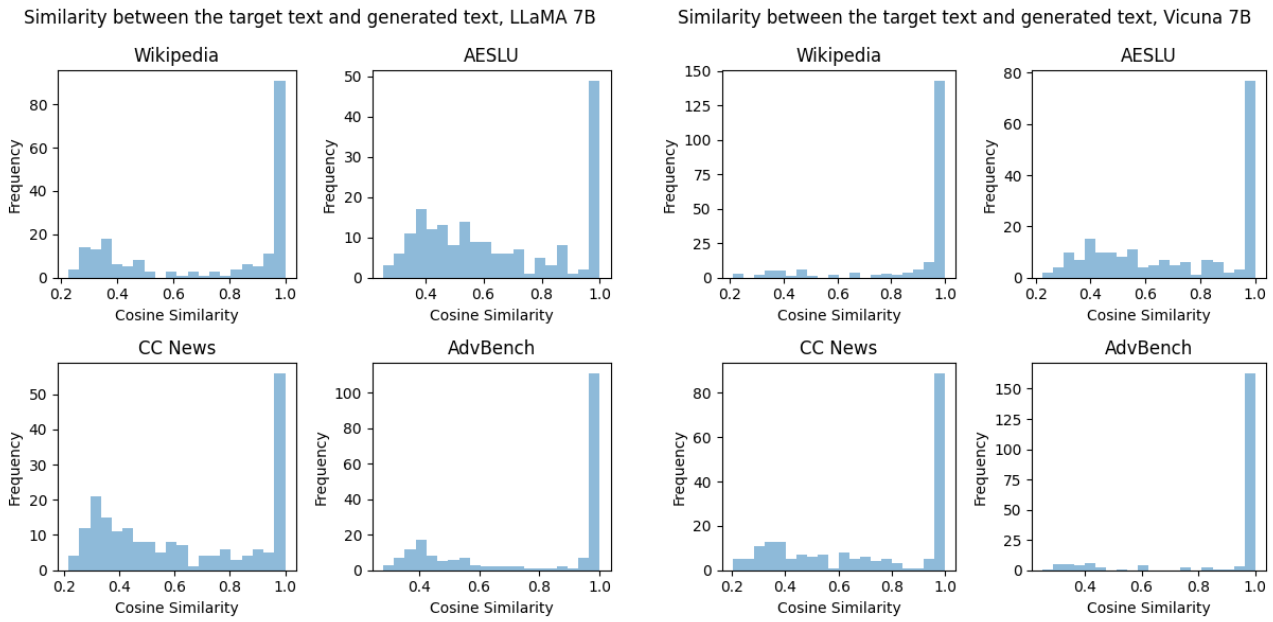

Figure 14. Histograms for distance between the target texts and model generations to Babel prompts across all datasets for 7B models.

B.9. Target Texts for Harry Potter Experiments

In Table 9, we include target texts containing factual information about the Harry Potter universe. Some of these texts were collected from the Wikipedia page devoted to the Harry Potter series[1], while others are responses to common trivia questions.

B.10. Hardware

All experiments were conducted using A100 GPUs. Computing Babel prompts for one model and one dataset required roughly 100 GPU hours, totaling approximately 1600 hours for all experiments.

[1] https://en.wikipedia.org/wiki/Harry_Potter

Table 9. Harry Potter Target Texts

Target Texts
Harry Potter is the main protagonist in J.K. Rowling's series of fantasy novels
Harry Potter's two best friends are Ron Weasley and Hermione Granger
Ron and Hermione went to the Gryffindor common room where they found Harry
Harry Potter is a series of seven fantasy novels
Hogwarts School of Witchcraft and Wizardry
Harry Potter and the Philosopher's Stone
Lord Voldemort is a dark wizard who intends to become immortal
Harry becomes a student at Hogwarts and is sorted into Gryffindor House
The trio develop an enmity with the rich pure-blood student Draco Malfoy
Defence Against the Dark Arts
Harry Potter and the Chamber of Secrets
Harry Potter and the Prisoner of Azkaban
Remus Lupin a new professor who teaches Harry the Patronus charm
Harry Potter and the Goblet of Fire
Hogwarts hosts the Triwizard Tournament
The Ministry of Magic refuses to believe that Voldemort has returned
Dumbledore re-activates the Order of the Phoenix
Snape teaches Defence Against the Dark Arts
Harry and Dumbledore travel to a distant lake to destroy a Horcrux
Lord Voldemort gains control of the Ministry of Magic
Harry Ron and Hermione learn about the Deathly Hallows
J. K. Rowling is a British author and philanthropist
Dobby is the house elf who warns Harry Potter against returning to Hogwarts
Hogwarts Durmstrang and Beauxbatons compete in the Triwizard Tournament
Minerva McGonagall is a professor of Transfiguration
Harry Potter's hometown is Godric's Hollow
Harry uses the fang of a basilisk to destroy Tom Riddle's diary" There 7 players on a Quidditch team
Hermione Granger's middle name is Jean
Nagini is the name of Voldemort's beloved snake
The Hogwarts Express departs from platform 9 3/4
Ollivanders is the name of the wand shop in Diagon Alley
Disarming Charm (Impedimenta)
Harry Potter's middle name is James
Harry's first crush is Cho Chang
The Weasley's house is called "The Burrow"
Hermione's cat is a white cat named Snowy
Albus Percival Wulfric Brian Dumbledore
His parents are named Lucinda and Cygnus Malfoy
Luna father's name is Xenophilius Lovegood
The four ghosts of Hogwarts houses are:
Ron has three siblings: Fred Weasley
The spell used to light the end of a wand is "Lumos"
Harry Potter and Lord Voldemort share a magical talent
Harry's muggle aunt and uncle are Marge and Vernon Dursley
Harry catches his first snitch in his fifth year at Hogwarts
James Potter Sirius Black Remus Lupin and Peter Pettigrew
The three Unforgivable Curses in the Harry Potter series are
Harry's wand is made of oak wood with a phoenix feather core
The Sorting Hat is a magical hat in the Harry Potter series that is used to
Albus Dumbledore has two siblings a brother and a sister. Their names are
Harry Potter is a talented Quidditch player who plays the position of Seeker
The Sorcerer's Stone is a powerful magical object from the Harry Potter series
1. Eye of newt 2. Wing of bat 3. Tail of lizard 4. Fang of toad 5. Snake's fang
Aragog is a fictional creature from the Harry Potter series by J.K. Rowling
The Half-Blood Prince is a character in the Harry Potter series by J.K. Rowling
Severus Snape is a potions master at Hogwarts School of Witchcraft and Wizardry
The goblin who helps Harry Ron and Hermione break into Gringotts is named Griphook
Harry Potter belongs to Gryffindor house at Hogwarts School of Witchcraft and Wizardry
Harry first hears the prophecy about himself in "The Philosopher's Stone" by J.K. Rowling
Harry gets the lightning bolt scar on his forehead as a result of Voldemort's attempt to kill him